

lab10(SSU)

October 12, 2025

1 Lab 10: Applying Functions and Iteration

Please complete this notebook by filling in the cells provided. Before you begin, execute the previous cell to load the provided tests.

Helpful Resource: - [Python Reference](#): Cheat sheet of helpful array & table methods!

Recommended Readings:

- [Tabular Thinking Guide](#)
- [Applying Functions](#)
- [Conditionals](#)
- [Iteration](#)

Please complete this notebook by filling in the cells provided. **Before you begin, execute the cell below to setup the notebook by importing some helpful libraries.** Each time you start your server, you will need to execute this cell again.

1.1 1. 2023 Cal Football Season

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install py pandoc
!pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
```

Suggested packages:

```
fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc
```

The following NEW packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexlua53 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils
```

0 upgraded, 54 newly installed, 0 to remove and 38 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1> [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-lato all 2.0-2.1> [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 poppler-data all 0.4.11-1> [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu/jammy/universe amd64 tex-common all 6.17> [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-urw-base35 all 20200910-1> [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13> [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1> [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 libijs-0.35 amd64 0.35-15build2> [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 libjbig2dec0 amd64 0.19-3build2> [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13> [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libkpathsea6>

amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
 Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
 1.0.2-1build4 [45.2 kB]
 Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
 2.13.1-1 [1,221 kB]
 Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
 2.004.5-6.1 [4,532 kB]
 Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
 20201225-1build1 [397 kB]
 Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
 20180621-3.1 [10.2 MB]
 Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
 all 18-1 [4,720 B]
 Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
 java all 43-1 [10.8 kB]
 Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
 java all 1.2-2 [60.3 kB]
 Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1
 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 http://archive.ubuntu.com/ubuntu jammy/main amd64 rubygems-integration
 all 1.18 [5,336 B]
 Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby3.0 amd64
 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
 all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
 [5,100 B]
 Get:25 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:26 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsynchronet2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all

```

1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 7s (25.5 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126675 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.

```

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Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...

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Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...

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Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack .../33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package tlutils.
Preparing to unpack .../38-tlutils_1.41-4build2_amd64.deb ...
Unpacking tlutils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.

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Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...

```



```

Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nm1ubuntu1) ...

```

Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
This may take some time... done.
Collecting py pandoc
Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)
Downloading py pandoc-1.15-py3-none-any.whl (21 kB)
Installing collected packages: py pandoc
Successfully installed py pandoc-1.15
Collecting datascience
Downloading datascience-0.18.0-py3-none-any.whl.metadata (910 bytes)
Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from datascience) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from datascience) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.2)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from datascience) (0.8.2)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Collecting jedi>=0.16 (from ipython->datascience)

 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)
Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)
Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (2.19.2)
Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.0)
Requirement already satisfied: matplotlib-inline in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience) (0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in /usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience) (0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (2025.10.5)
Downloading datascience-0.18.0-py3-none-any.whl (725 kB)

725.0/725.0 kB

12.8 MB/s eta 0:00:00

Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)

1.6/1.6 MB

50.0 MB/s eta 0:00:00

Installing collected packages: jedi, datascience

Successfully installed datascience-0.18.0 jedi-0.19.2

```
[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Run this cell to set up the notebook, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
import warnings
warnings.simplefilter('ignore', FutureWarning)
```

Jonathan and Jadyn are trying to analyze how well the Cal football team performed in the 2023 season. A football game is divided into four periods, called quarters. The number of points Cal scored in each quarter and the number of points their opponent scored in each quarter are stored in a table called `cal_fb.csv`.

```
[4]: # Just run this cell
# Read in the cal_fb csv file
games = Table().read_table("./DS/cal_fb.csv")
games.show()
```

<IPython.core.display.HTML object>

Let's start by finding the total points each team scored in a game.

Question 1. Write a function called `sum_scores`. It should take four arguments, where each argument represents integers corresponding to the team's score for each quarter. It should return the team's total score for that game. **(2 Points)**

Hint: Don't overthink this question!

```
[6]: def sum_scores(q1, q2, q3, q4):
      '''Returns the total score calculated by adding up the score of each
      ↪quarter'''
      return q1 + q2 + q3 + q4

      sum_scores(14, 7, 3, 0) #DO NOT CHANGE THIS LINE
```

[6]: 24

Question 2. Create a new table `final_scores` with three columns in this *specific* order: Opponent, Cal Score, Opponent Score. You will have to create the Cal Score and Opponent Score columns. Use the function `sum_scores` you just defined in the previous question for this problem. (5 Points)

Hint: If you want to apply a function that takes in multiple arguments, you can pass multiple column names as arguments in `tbl.apply()`. The column values will be passed into the corresponding arguments of the function. Take a look at the Python Reference Sheet and Lecture 13's demo for syntax.

Note: If you're running into issues creating `final_scores`, check that `cal_scores` and `opp_scores` output what you want. If you're encountering `TypeError`s, check the [Python Reference](#) to see if the inputs/outputs of the function are what you expect.

```
[8]: cal_scores = games.apply(sum_scores, "Cal 1Q", "Cal 2Q", "Cal 3Q", "Cal 4Q")
      opp_scores = games.apply(sum_scores, "Opp 1Q", "Opp 2Q", "Opp 3Q", "Opp 4Q")
      final_scores = Table().with_columns(
          "Opponent", games.column("Opponent"),
          "Cal Score", cal_scores,
          "Opponent Score", opp_scores
      )
      final_scores
```

```
[8]: Opponent      | Cal Score | Opponent Score
      Nevada       | 17        | 22
      TCU           | 32        | 34
      Sacramento State | 42        | 30
      Washington    | 24        | 31
      Washington State | 6         | 21
      Oregon        | 17        | 24
      Colorado      | 26        | 3
      Oregon State   | 39        | 25
      Arizona       | 3         | 10
      Stanford      | 41        | 11
      ... (2 rows omitted)
```

We can get specific row objects from a table. You can use `tbl.row(n)` to get the `n`th row of a table. `row.item("column_name")` will allow you to select the element that corresponds to `column_name` in a particular row. Here's an example:

```
[9]: # Just run this cell
# We got the Axe!
games.row(9) # <-- this will return a row object
```

```
[9]: Row(Opponent='Stanford', Cal 1Q=0, Cal 2Q=14, Cal 3Q=13, Cal 4Q=14, Opp 1Q=0,
Opp 2Q=3, Opp 3Q=0, Opp 4Q=8)
```

```
[10]: # Just run this cell
games.row(9).item("Cal 4Q") # <-- this will return a item (e.g. an int) from a
↳row object
```

```
[10]: 14
```

Question 3. We want to see for a particular game whether or not Cal lost. Write a function called `did_cal_lose`. It should take one argument: a **row object** from the `final_scores` table. It should return either `True` if Cal's score was less than the Opponent's score, and `False` otherwise. (5 Points)

Note 1: "Row object" means a row from the table extracted (behind the scenes) using `tbl.row(index)` that contains all the data for that specific row. It is **not** the index of a row. Do not try and call `final_scores.row(row)` inside of the function.

Note 2: If you're still confused by row objects, try printing out `final_scores.row(1)` in a new cell to visually see what it looks like! This piece of code is pulling out the row object located at index 1 of the `final_scores` table and returning it. When you display it in a cell, you'll see that it is not located within a table, but is instead a standalone row object!

Hint: You can use comparators like `<`, `>`, `<=`, `>=`, `==`, and `!=` to compare items from a row object

```
[12]: def did_cal_lose(row):
      return row.item("Cal Score") < row.item("Opponent Score")

did_cal_lose(final_scores.row(1)) #DO NOT CHANGE THIS LINE
```

```
[12]: True
```

Question 4. Jonathan and Jadyne wants to see how Cal did against every opponent during the 2023 season. Using the `final_scores` table:

1. Assign `results_array` to an array of `True` and `False` values that correspond to whether or not Cal lost.
2. Add `results_array` to the `final_scores` table in a column named `Results`, and assign this to `final_scores_with_results`.
3. Then, respectively assign the number of wins and losses Cal had to `cal_wins` and `cal_losses`.

(10 Points)

Hint: `tbl.apply()` might be helpful. Refer to the [Python Reference](#) if you're unsure how it works!

```
[14]: results_array = final_scores.apply(did_cal_lose)
final_scores_with_results = final_scores.with_column("Results", results_array)
cal_losses = results_array.sum()
cal_wins = len(results_array) - cal_losses

# Don't delete or edit the following line:
print(f"In the 2023 Season, Cal Football won {cal_wins} games and lost_
↳{cal_losses} games. Go Bears! ")
```

In the 2023 Season, Cal Football won 5 games and lost 7 games. Go Bears!

1.2 2. Unrolling Loops

“Unrolling” a for loop means to manually write out all the code that it executes. The result is code that does the same thing as the loop, but without the structure of the loop. For example, for the following loop:

```
for num in np.arange(3):
    print("The number is", num)
```

The unrolled version would look like this:

```
print("The number is", 0)
print("The number is", 1)
print("The number is", 2)
```

Unrolling a for loop is a great way to understand what the loop is doing during each step. In this exercise, you’ll practice unrolling a for loop.

In the question below, write code that does the same thing as the given code, but with any for loops unrolled. It’s a good idea to run both your answer and the original code to verify that they do the same thing. (Of course, if the code does something random, you’ll get a different random outcome than the original code!)

Question 1. Unroll the code below.

```
[15]: for joke_iteration in np.arange(3):
    print("Knock. " * (joke_iteration + 1))
    print("Who's there?")
    print("Banana.")
    print("Banana who?")
print("Knock, knock.")
print("Who's there?")
print("Orange.")
print("Orange who?")
print("Orange you glad I didn't say banana?")
```

```
Knock.
Who's there?
Banana.
Banana who?
```


Knock. Knock.
Who's there?
Banana.
Banana who?
Knock. Knock. Knock.
Who's there?
Banana.
Banana who?
Knock, knock.
Who's there?
Orange.
Orange who?
Orange you glad I didn't say banana?

```
[16]: print("Knock. ")  
      print("Who's there?")  
      print("Banana.")  
      print("Banana who?")  
      print("Knock. Knock. ")  
      print("Who's there?")  
      print("Banana.")  
      print("Banana who?")  
      print("Knock. Knock. Knock. ")  
      print("Who's there?")  
      print("Banana.")  
      print("Banana who?")  
      print("Knock, knock.")  
      print("Who's there?")  
      print("Orange.")  
      print("Orange who?")  
      print("Orange you glad I didn't say banana?")
```

Knock.
Who's there?
Banana.
Banana who?
Knock. Knock.
Who's there?
Banana.
Banana who?
Knock. Knock. Knock.
Who's there?
Banana.
Banana who?
Knock, knock.
Who's there?
Orange.

```
Orange who?  
Orange you glad I didn't say banana?
```

```
Knock.  
Who's there?  
Banana.  
Banana who?  
Knock. Knock.  
Who's there?  
Banana.  
Banana who?  
Knock. Knock. Knock.  
Who's there?  
Banana.  
Banana who?  
Knock, knock.  
Who's there?  
Orange.  
Orange who?  
Orange you glad I didn't say banana?
```

1.3 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[ ]: # should change the directory and file name matching to yours  
!jupyter nbconvert './DS/lab10(SSU).ipynb' --to pdf
```